

New type of primers based on polypropylene grafted maleic anhydride (PP-g-MAH) –strength analysis

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ABSTRACT: Primers are an appealing approach for enhancing adhesion to low-energy surfaces owing to their cost-effectiveness in manufacturing and operation, high efficiency, and scalability. Commercial adhesion coatings typically contain 3-20% chlorinated polyolefins, which are environmentally unstable and contribute to the overall cost. This study proposes primers based on polypropylene grafted with maleic anhydride (PP-g-MAH) as a novel alternative. The research investigates the strength of adhesive joints formed using a PP-g-MAH-based primer. Polypropylene substrates coated with the primer were bonded to polyethylene foam coated with acrylic adhesive. The study examines various factors, including concentration

of PP-g-MAH in the primer solution, the degree of grafting of polypropylene with maleic anhydride, the solvent type, the temperature of primer application, and pressure duration and magnitude. Peel strength analysis demonstrates that PP-g-MAH primers significantly enhance the adhesion of polypropylene substrates to the acrylic adhesive on the adhesive tape, requiring only a 1% solution to achieve maximum joint strength (averaging 35.53 N and 37.29 N at 0.5% and 1% concentrations, respectively). Moreover, peel strength decreases with increasing polypropylene modification with maleic anhydride (41.1 N for 0.66% grafting, 31.1 N for 1.0%, and 29.7 N for 1.43%). High-temperature application is necessary, with peak strength observed at 100 °C (54.4% increase compared to 25 °C) and 120 °C (57.9%). However, solvent selection can influence primer application temperature, with decalin and toluene yielding the highest strength (averaging 45.2 N and 42.8 N) compared to tetrachloroethane and xylene (37.3 N and 35.5 N). Pressure application is crucial for bonding polymers with foam adhesive tape, significantly enhancing strength (e.g., 71% increase at 20 bar) with short-duration pressure (42.9 N for 10 seconds, 33.5 N for 60 seconds).

1. Introduction

Polypropylene, widely used in many industries¹, is characterized by repeatable hydrogen-carbon units in the chemical structure, resulting in a low surface free energy of 29 to 35 mJ/m². This property gives chemical resistance to items made of polypropylene. On the other hand, it causes major technological problems in surface treatment related to painting and bonding².

Several methods have been used in industrial practice to increase the surface adhesion of polypropylene: physical melting, flame and plasma activation, corona discharge, high-energy radiation, or immersion in a chromium mixture³⁻⁹. Primers (adhesion coatings) are also a widespread method for increasing the adhesion of polypropylene substrates. They are polymer

solutions that, when applied to a surface, combine through the diffusion mechanism with a hardly wettable substrate and then with the applied layers of paint or adhesive. Adhesion coatings applied to the polypropylene surface leave a transparent layer that does not damage the substrate, and, in contrast, they are often responsible for increased resistance to weathering and aging^{10,11}.

Adhesive coatings have become permanent fixtures in production practice because of their high efficiency, low cost, and use on any production scale alone or with other surface activation methods. It is prevalent in the plastic car parts supply industry.

Several groups of substances can be used to make primers for polyolefin surfaces. Among them are alkylamines, transition metal complexes, dopamine-based proteins, alkylboranes, and functionalized polyolefins¹²⁻¹⁷.

As a result of the simple preparation and low production costs, the last listed substances quickly gained wide popularity in the composition of adhesive primers for plastic and metal surfaces. Polyolefins, mostly polypropylene, are modified with various substances, e.g., chlorine gas, acrylic acid, styrene, glycidyl methacrylate, phthalic anhydride, and maleic anhydride¹⁸⁻²¹. Among them, the most important is chlorinated polypropylene, obtained by reacting gaseous chlorine and polypropylene in a solution or suspension. Other methods of obtaining this substance have also been developed. However, they are more expensive and less effective^{22,23}.

Polypropylene used in primers has a low average molecular weight (15-30.000 g/mol) and exhibits various adhesive properties depending on the degree of chlorination. Higher crystallinity and better adhesive properties, but limited solubility characterizes less chlorinated polypropylene (20% and lower). On the contrary, chlorinated polypropylene in the 20-30% range has lower crystallinity and solubility and is less effective in increasing adhesion properties. Primers

containing chlorinated polypropylene (CPO) have a concentration of 3-20% and are often combined with other polymer resins to increase adhesion^{17,19,24,25}. In addition, they are also grafted with maleic anhydride, which improves the compatibility of the polymer chain^{25,26}.

We propose maleic anhydride-grafted polypropylene (PP-g-MAH) as an alternative to chlorinated polypropylene. It is mainly used as an adhesion promoter additive in plastics. This application increases miscibility with other polar components of plastics, such as glass and carbon fibres, graphene, fillers, and other polymers or even with metals, e.g., aluminum²⁷⁻³¹. Until now, PP-g-MAH has not been used in polyolefin primers.

Currently, a popular trend is to minimize harmful substances that can enter the environment. The production and use of materials based on chlorinated polyolefins pollute the natural environment with chlorine. European Union regulations discourage manufacturing products containing chlorinated polyolefins, which are therefore subject to a high duty due to their harmful nature³². Thus, primers based on polypropylene modified only with maleic anhydride could reduce the amount of toxic waste. For this reason, it is reasonable to look for a replacement of chlorinated polyolefins with the primer composition.

Novak and co-workers showed that by modifying polypropylene with maleic anhydride, the bonding efficiency of aluminum increases³³, while CPOs are mainly suitable for bonding polyolefins³⁴. This suggests that PP-g-MAH in primer form may also be effective against other substrates, not only polyolefin ones.

Williams, Germinario, and Eagan³⁵ showed that non-chlorinated primers give high initial adhesion and a layer with excellent resistance to gasoline. Furthermore, mixing them with CPO does not result in a decrease in adhesion joint strength.

CPO-based primers are used in the system with adhesives and coatings based on acrylates, epoxy resins, polyurethanes and cyanoacrylates^{33,34}. Therefore, the use of PP-g-MAH as a primer base can lead to an expansion of the number of types of adhesives for polyolefin substrates to include more hydrophilic polyurethane and epoxy resin systems or adhesives filled with polar additives^{36,37}.

With these particular properties of PP-g-MAH considered, studying this material as a primer that acts on the surface rather than in the bulk of the polymer is crucial. This paper presents a strength analysis of adhesive joints created using PP-g-MAH primers and a solvent: effect of PP-g-MAH concentration, polypropylene grafting degree with maleic anhydride, influence of kind of solvent, primer application temperature, and power and time of pressure.

The research can shed light on new applications for polypropylene grafted with maleic anhydride, which is popular in the industry, and can reduce the amount of harmful chlorine released into the environment. The primer mixture based on non-chlorinated substances has not been known in the industry and is currently in the patent process³⁸.

2. Materials

Polypropylene strips (20 x 203 x 2 mm) from Hostacom TRC 352N E1 C11306 (Melt Rate MFR = 18g / 10 min (230 °C / 2.16 kg), $d = 1.05\text{g} / \text{cm}^3$ (23 °C), (Basell Orlen Polyolefins, Poland) were used as the substrate.

The substrates were cleaned with isopropanol 99.7% (Alchem, Poland). Primers were prepared by dissolving polypropylene homopolymer modified with maleic anhydride from the Bondyram series (Polyram, Israel), shown in Table 1, using a xylene isomers mixture of 98.5% (Alchem, Poland).

Table 1. Materials used in primer compositions.

Material	Maleic Anhydride Content [%] FTIR	MFR(190/2.16 kg) ISO 1133 (g/10 min)
Bondyram 1001-LE	0.66	75.80
Bondyram 1001	1.00	105.67
Bondyram 2101	1.43	192.00

Double-sided tape DuploCOLL 56066 (Lohmann, Poland) (Shear strength (23 °C) = 50 N/625 mm², peel strength (23 °C) = 25 N/25 mm) with a width of 20 mm was used for the peel 180 strength test according to ASTM D903 – 98. The tape consists of polyethylene foam with a thickness of 3.20 mm coated with acrylic adhesive.

All parts in the adhesive joint (substrate, adhesive tape, and primer) are commonly used in the automotive industry.

When forming adhesive bonds, liquid adhesives are usually used³⁹, which ensure high strength but do not allow for obtaining a flexible and durable joint simultaneously. This study used foam adhesive tape coated with a thin layer of acrylic adhesive. Foam adhesive tape compensates for the internal stresses of the joint when joining uneven surfaces and damp vibrating components. For this reason, foam tape is commonly used to connect separating walls with glass panes in the automotive industry.

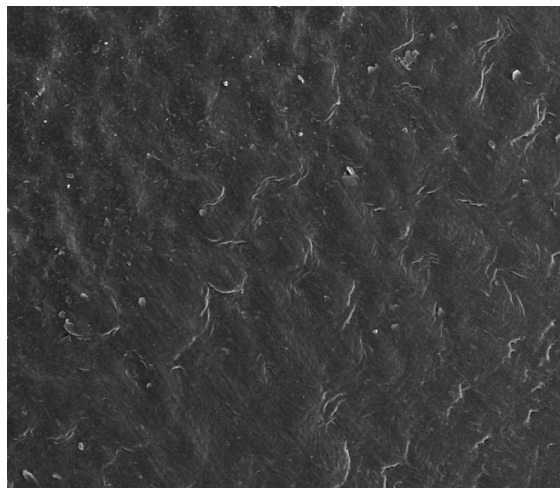
2.1. Samples preparation

A detailed flow chart of the study is attached to supporting information. The test substrates were obtained from the original car parts for Škoda Superb produced by Maflow Plastics Poland, which are parts of the boot. These parts were injected from polypropylene Hostacom TRC 352N

E1 C11306. They have a superimposed K3N car texture according to the VW classification, characterized by fine graining.

The primers were obtained by dissolving the appropriate amount of Bondyram series maleic anhydride grafted polypropylene. The solutions obtained were almost clear at low concentrations (<1%) and cloudy at higher concentrations. Polymer solutions at room temperature, especially at high concentrations (>5%) tended to gel.

Polypropylene strips (20 x 203 x 2 mm) (Figure 1) were washed with isopropanol and covered with a hot (100 °C) 0.5% xylene solution of the primer containing one of the polymers from the Bondyram series unless otherwise stated. After evaporating the solvent, the substrate was joined with DuploCOLL 56005 adhesive tape using a hydraulic press (BegoHydrofix, Germany). DuploCOLL 56005 is polyethylene foam-coated acrylic glue. A standard pressure of 20 bar and a holding time of 60 seconds were used for joining unless otherwise stated. The test specimens were conditioned at $50\% \pm 5\%$ humidity and $23\text{ °C} \pm 1\text{ °C}$ for 30 days before testing. The same conditions prevailed during the strength tests in the accredited laboratory.



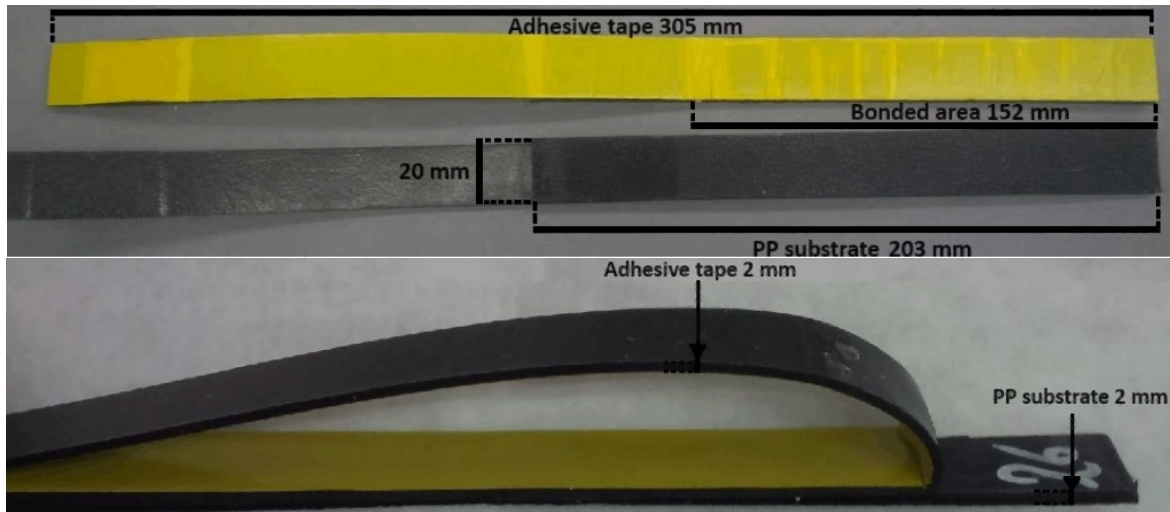


Figure 1. Up – automotive fine graining at x150 resolutions; down - appearance and dimensions of specimens

There are two most popular strength tests: lap shear⁴⁰ and peel strength⁴¹. Peel strength 180 was selected in the study because peeling is a more common phenomenon in the plastic automotive parts supplier industry, e.g., pushing out glass pane from plastic separating wall in the car.

Mechanical properties were measured at room temperature using a TIRA 27025 test machine (TIRA Maschinenbau GmbH, Germany). The measurements followed ASTM D903 – 98 standards at 152 mm/min velocity.

Three repetitions of each test were performed. Cohesive failure of the adhesive tape was observed for joints with strengths higher than 45 N. The maximum peel strength achieved in the test was taken to calculate the average (Figure 2).

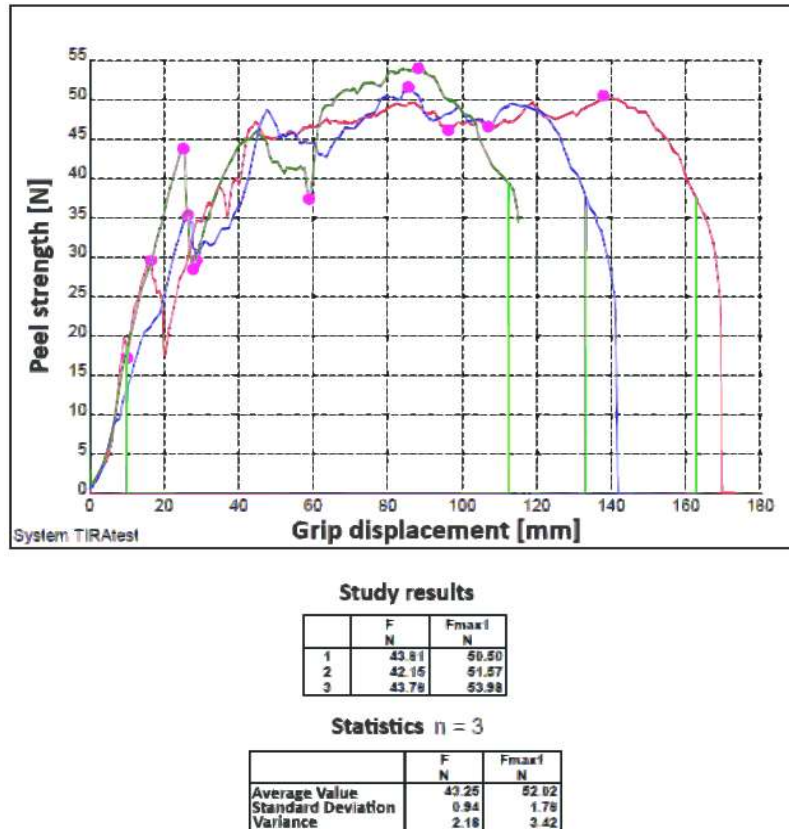


Figure 2. Peel strength test results from Bondyram 1001-LE sample at 1% concentration

3. Results and discussion

The following parameters were investigated for their effect on the peeling strength of adhesive joints composed of polypropylene substrate, primer layer, and adhesive tape:

- primer concentration,
- the temperature of primer application,
- type of solvent,
- time and pressure of the hydraulic press,
- grafting degree of polypropylene with maleic anhydride.

3.1. PP-g-MAH stability under primer preparation conditions

Commercially available polypropylene grafted maleic anhydride contains simultaneously both anhydride and acid groups in its structure. Therefore, it is essential to exclude the possibility of succinic anhydride hydrolysis to maintain constant test conditions.

Sclavons et al.⁴² report that this substance does not yield hydrolysis under these conditions. We can confirm this information after obtaining the FTIR spectra of raw polymers precipitated from primers (Figure 3). The succinic anhydride hydrolysis reaction should cause a significant increase in the intensity of the carbonyl peak 1710-1720 cm^{-1} and the disappearance of the anhydride band 1780-1790 cm^{-1} .

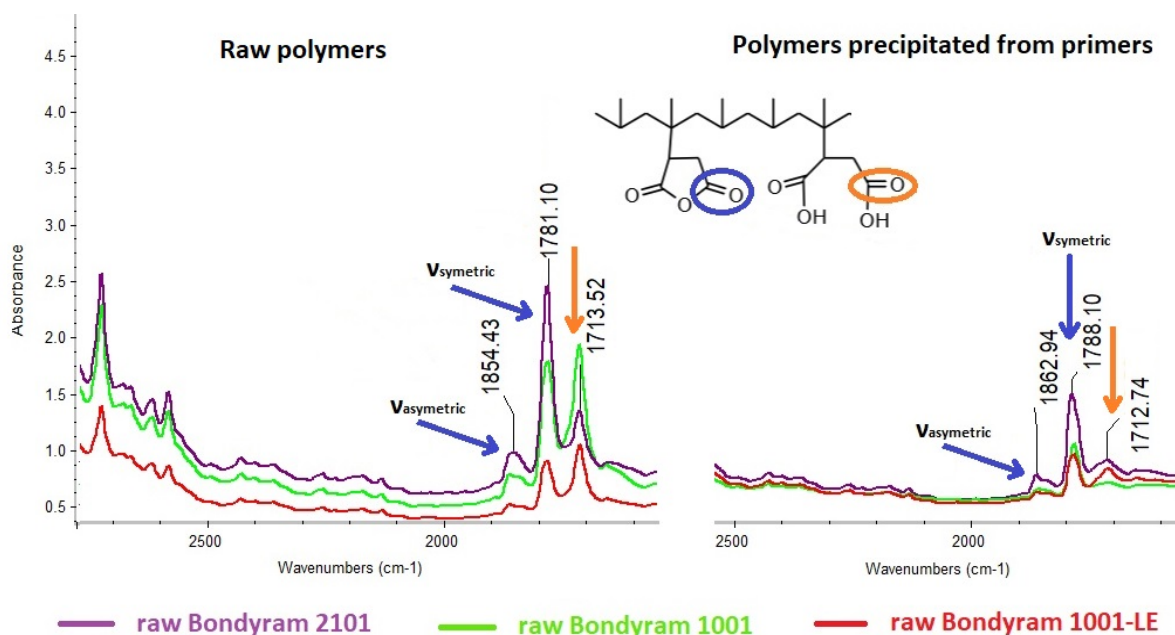


Figure 3. FTIR spectra of raw and precipitated from primers Bondyram series PP-g-MAH.

Hydrolysis can occur, for example, in the presence of pyridine, which catalyzes this reaction, or by using water in a strongly acidic environment. However, due to the polymer's hydrophobic nature and chemical resistance, even these conditions may not result in hydrolysis.

3.2. Primer concentration

This study examined the effect of increasing the concentration of PP-g-MAH in the primer solution on the peel strength of the adhesive joints. Xylene solutions of individual polymers (Table 1) were prepared with increasing concentrations: 0.1, 0.5, 1.0, 5, and 10%. Other joint preparation parameters, such as primer application temperature, time, and pressure force are maintained as described in the section *Samples Preparation*. The obtained strength results are shown in Figure 4.

Commercial primers use chlorinated polyolefins, most commonly chlorinated polypropylene. The concentration of this substance in the adhesive primer solution is usually high- between 3 and 20%, and the peeling strength of the joints formed after primer application increases proportionally with the concentration of chlorinated polypropylene⁴³.

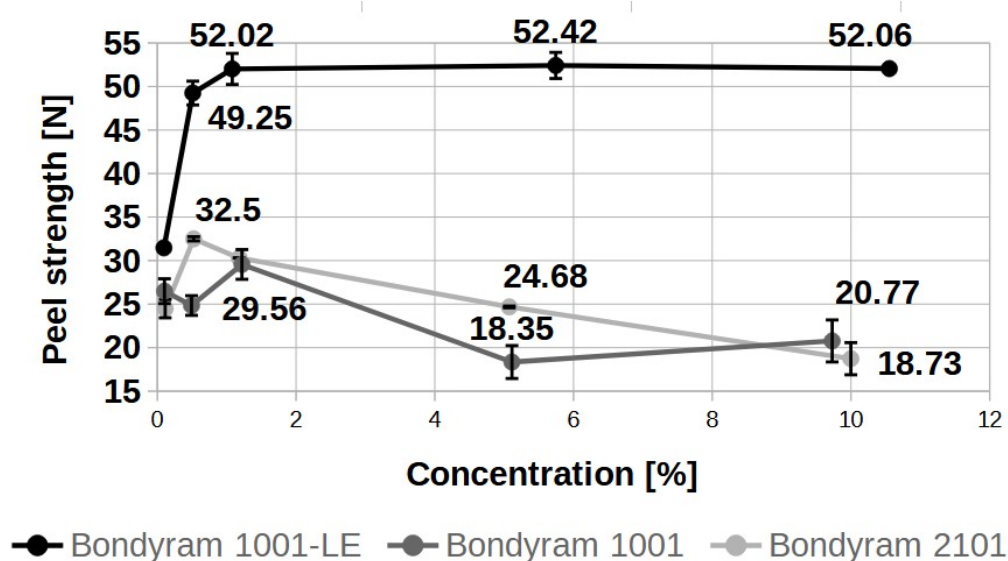


Figure 4. Influence of concentration of the primer solution on peel strength.

In our study, the maximum peel strength was achieved using a concentration of only 0.5% (average 35.53 N). At this concentration, primer solutions were almost transparent. At a concentration of 1%, the strength of the samples was also high- averaged 37.29 N for all polymers. However, after exceeding a concentration of 0.5%, an increase in the cloudiness of all

solutions was observed. Above 5%, all xylene solutions started to gel after decreasing the temperature of the primer below 130 °C. In the case of Bondyram 1001-LE, with the lowest degree of grafting, despite the gelation of the primer, the peel strength was still maintained at the level of the cohesive strength of the adhesive tape. For the remaining polymers above a concentration of 5%, the strength of the joints decreased dramatically.

Solutions containing less than 1% polypropylene grafted with maleic anhydride after application gave a transparent layer. However, the adhesive primer was visible above this concentration as a white layer (Figure 5). Furthermore, after the peeling tests, the white layer remained on the substrate, not the adhesive tape. This means there is insufficient adhesion between the primer layer and the adhesive tape, compared to the substrate and the primer layer.

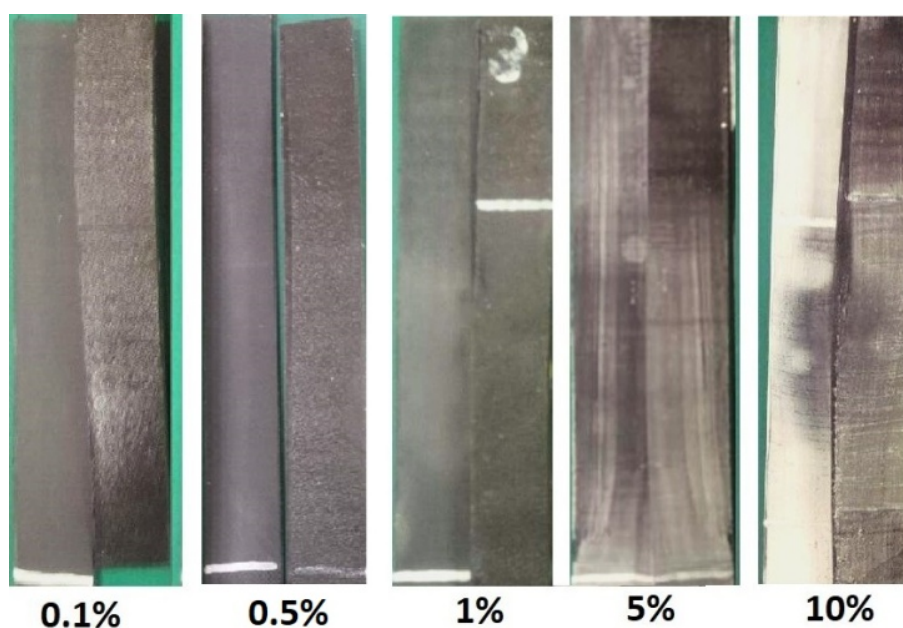


Figure 5. Layers of primer based on Bondyram 2101 with increasing concentration. On the left of each photo is a polypropylene substrate, and on the right is a detached adhesive tape.

Strong nucleation of the concentrated primer is caused by the presence of maleic anhydride in the polypropylene structure, which reduces the softening point of the polymer and thus increases susceptibility to renucleation⁴⁴. It is well known that the adhesion of adhesives and primers depends largely on viscosity, the increase of which limits the diffusion of polymer chains into the substrate and the formation of interphases^{45,46}. The rise in viscosity also results in applying a thicker and less even layer.

3.3. The temperature of primer application

In the presented studies, the primer (0.5% xylene solution of each polypropylene grafted maleic anhydride from Table 1) was applied at various temperatures: 120, 100, 80, 50, and 25 °C. The remaining parameters (time and holding pressure) were maintained according to the section *Samples Preparation*. The strength results obtained are presented in Figure 6.

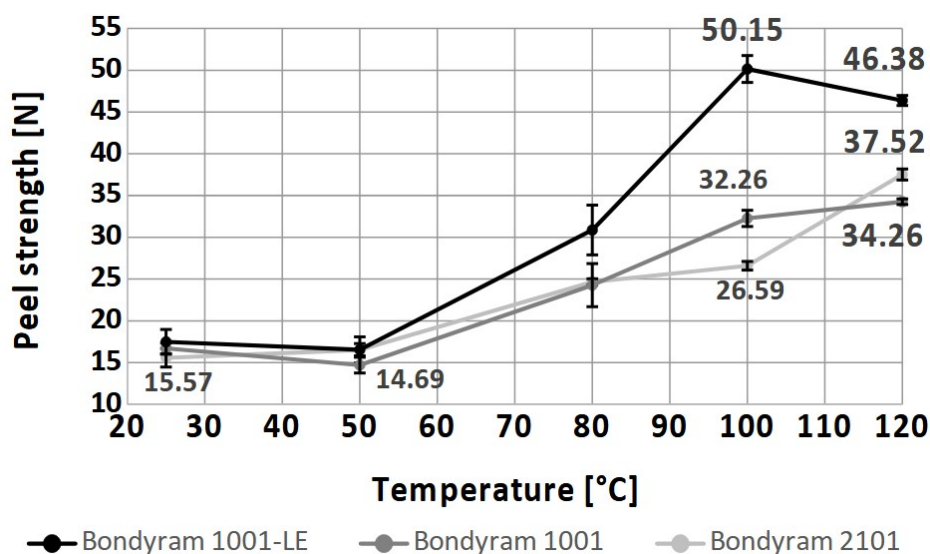


Figure 6. Dependence of the primer application temperature on the maximum peel strength.

A similar trend has been observed for all polymers tested: at low primer application temperatures reaching 50 °C, the peel strength is constant and remains very low. An increase in strength is apparent when the solution became more fluid at 80 °C (average 26.6 N). The

durability of the joint above 80 °C increases proportionally with temperature to the maximum value at the highest primer application temperature tested at 120 °C (average 39.39 N). The effect of the increased strength of the samples was evident in the form of more extensive areas of adhesive tape remaining on the PP substrates after the following tests (Figure 7). In the low range of primer applying temperature below 50 °C, the primer was in the form of a gel, which, as the temperature increased, changed its consistency and viscosity into the liquid.

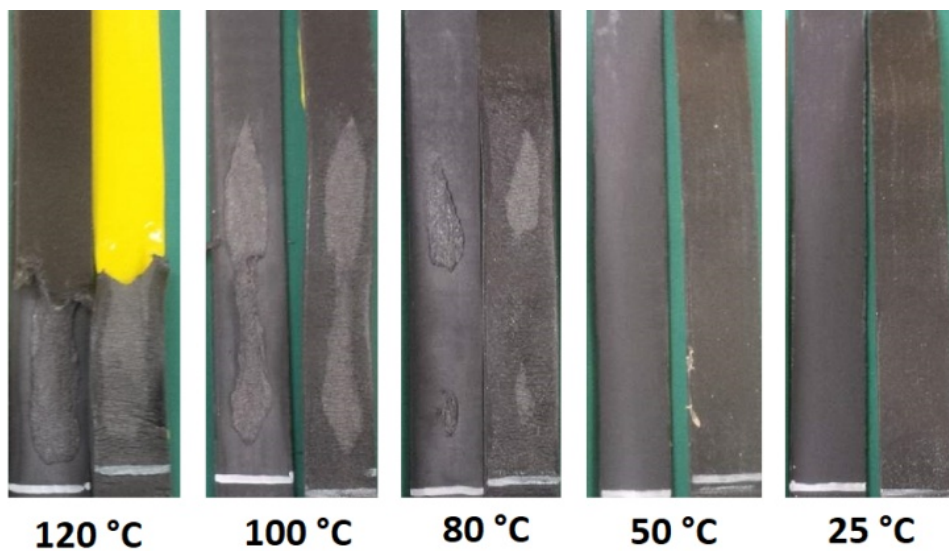


Figure 7. Change in the appearance of the samples with increasing primer application temperature (Bondyram 1001-LE). The right side of each image is polypropylene substrate, and the left side is detached adhesive tape.

The polymer chains are rigid and highly crystallized at low temperatures, arranged next to each other. An increase in temperature and the presence of solvent causes the stiff crystalline regions to loosen, the polymer chains to become flexible, and the intermolecular movements increase. Under these conditions, PP-g-MAH migrates much more easily into the substrate, as it does not encounter geometrical obstacles in its migration path. After cooling and evaporation of the solvent, PP-g-MAH forms an interphase, intertwining with the substrate matrix^{45,46}. The

dissolution of the polypropylene at low temperatures is inefficient, so the primer does not penetrate the substrate or create an interphase to increase peel strength⁴⁷.

3.4. Type of solvent

Primers containing 0.5% of each polymer (Table 1) were prepared using xylene, toluene, tetrachloroethane, and decalin as solvents. All polymers were dissolved in suitable solvents. Other parameters, such as application temperature, time, and pressing force, were applied in the section Sample Preparation. The obtained strength results are presented in Figure 8.

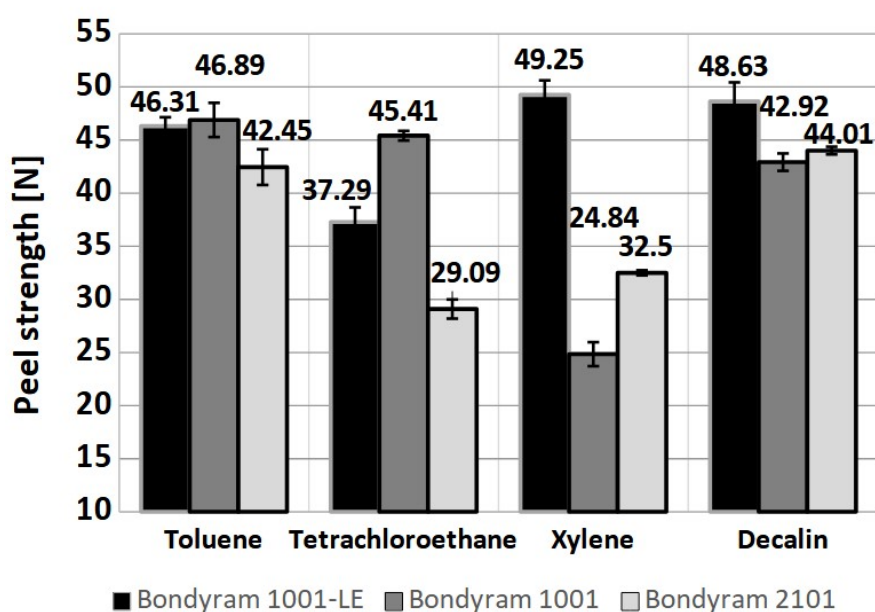


Figure 8. Effect of different solvents used in the primer on peel strength.

The peel strength values were averaged for each solvent and the following series was obtained: decalin (45.19 N) > toluene (42.79 N) > tetrachloroethane (37.27 N) > xylene (35.54 N).

As a result of the high boiling point and similar chemical structure, decalin easily dissolves the polypropylene substrate. Decalin is a saturated hydrocarbon composed of two fused cyclohexane rings that exist as two isomers, *trans*- and *cis*-decalin. Unlike *trans*-decalin, which has a rigid structure, *cis*-decalin is flexible and occurs as two conformers. Commercial decalin is a mixture

of 60:40 *trans*- and *cis*-isomer. Hydrocarbon structure allows it to fit geometrically and penetrate even into the densely packed polymer chains of the polypropylene substrate. In contrast to decalin, xylene has a rigid aromatic structure that cannot conform, and therefore penetration of the crystallized polymer structure is more difficult.

The primer obtained from modified polypropylene readily permeates the swollen surface. However, the solvent evaporates very slowly after application. Despite its good adhesive properties, it could be a better solvent in industry, where the duration of each process is essential.

Regardless of the polymer dissolved in toluene, stable-strength joints have been obtained. In contrast, despite a similar chemical structure and differing only in one methyl group, xylene provides joints of varying strength for the tested solvents. The increasing number of carbon atoms in the molecule makes it increasingly hydrophobic, reducing the number of polar interactions. Toluene, which has fewer carbon atoms than xylene, interacts with polar substances more readily and readily dissolves PP-g-MAH. However, the smaller molecule gives toluene greater mobility to migrate deeper into the polypropylene matrix. The differences between the hydrophilicity of toluene and xylene are shown by the Hansen Solubility Parameters (Table 2).

Table 2. Hansen Solubility Parameters of toluene and xylene⁴⁸

Solvent	dD^a	dP^b	dH^c
Toluene	18	1.4	2
Xylene	17.6	1	3.1
^a Dispersive Component, ^b Polar Component, ^c Hydrogen Bonds			

These values may indeed lead to an increase in the solubility of PP-g-MAH. Moreover, this solvent's lower share of hydrogen interactions leads to its increased mobility compared to xylene.

Tetrachloroethane was characterized by the highest volatility and the lowest solubility relative to PP-g-MAH. The high volatility makes the solvent suitable for industrial applications. However, the limited solubility of the active substance significantly reduces the strength of the joints.

3.5. Time and pressure of the hydraulic press

Primers were prepared at a concentration of 0.5% in xylene. They were applied at 100 °C and the samples were bonded with an adhesive tape under different hydraulic press pressures: 0, 20, 50, and 100 bar maintained for 60 seconds (Figure 9) and, in a further experiment, at a constant pressure of 20 bar and at different times of holding pressure: 10, 60, 120, 300 seconds (Figure 10).

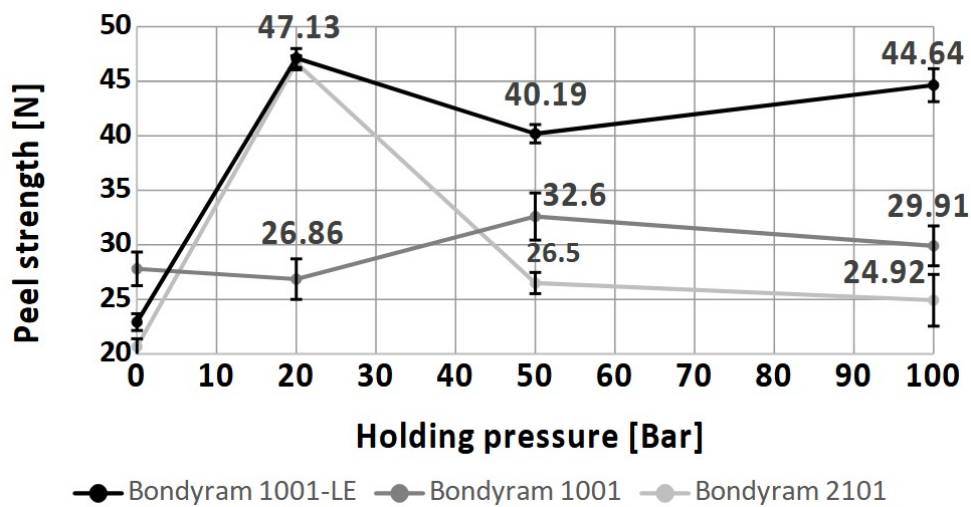


Figure 9. Effects of the hydraulic press pressure used to obtain adhesive joints on the peeling strength of the samples.

The adhesive bond is weak without applying any pressure (Figure 9). In all cases, the application of pressure strengthens the joint. The most significant reinforcement was observed at low pressure, while stronger resulted in decreased merger strength.

On the other hand, joints formed by applying pressure for a short time are characterized by high strength (Figure 10). Maintaining continuous pressure for a longer period of time will, in any case, reduce the power of the adhesive bond. With the increasing holding time, the peeling strength of all polymers decreases and remains constant.

Both studies show that foam adhesive tape should be kept from being compressed for too long time or too firm pressure. Due to high pressure, the tape is deformed and loses contact with the sample surface. The adhesive and the primer could be partially squeezed out from under the sample.

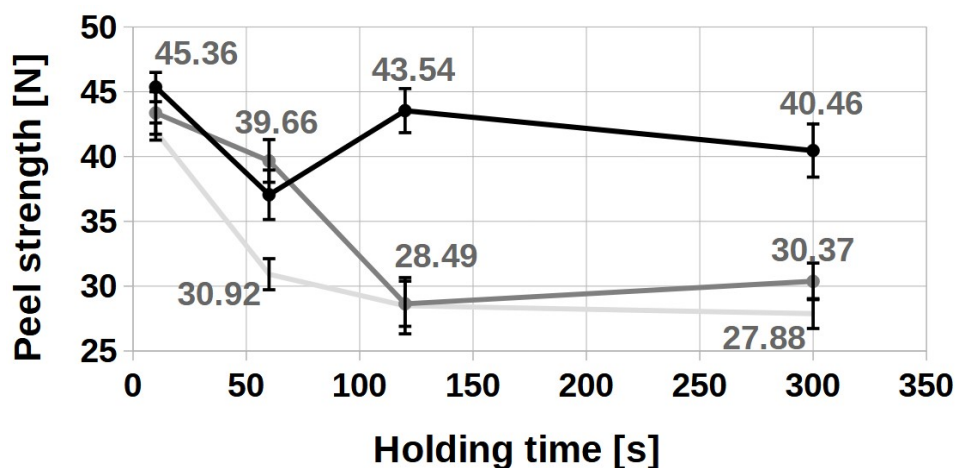


Figure 10. Variation in the peel strength of adhesive joints at different holding times of pressure of the hydraulic press.

3.6. Polypropylene grafting degree with maleic anhydride

Modifying polypropylene with maleic anhydride enhances the reactivity, compatibility, and polarity⁴⁹. Non-functionalized polypropylene is soluble in a few solvents with a high boiling point, such as xylene, toluene, decalin, and tetrachloroethane. However, unlike chlorine grafting of polypropylene, maleic anhydride functionalization reduces solubility of the polymer in poorly hydrogen-bonding solvents, for example, aromatic and cyclic hydrocarbons⁵⁰.

In the studies described above, three polymers with increasing degrees of functionalization with maleic anhydride were always compared: Bondyram 1001-LE 0.66%, Bondyram 1001 1.00%, and Bondyram 2101 1.43%. The conditions for each experiment were the same. Therefore, all the results obtained for a given polymer were averaged (Table 3), e.g., for Bondyram 1001-LE in the Concentration section, the results obtained for subsequent concentrations of 0.1, 0.5, 1, 5, and 10% of the polymer were averaged.

Table 3. Average values of all data obtained from experiments for each polymer with increasing maleic anhydride content.

Experiment	Bondyram 1001-LE 0.66% [N]	Bondyram 1001 1.00% [N]	Bondyram 2101 1.43% [N]
Concentration	47.44	24.00	26.14
Solvent	45.37	40.02	37.01
Temperature	32.28	24.44	24.17
Holding pressure	38.72	29.29	29.69
Holding time	41.49	37.63	31.41
Average	41.06	31.08	29.68

Strength tests indicate an increase in adhesion properties with decreasing degree of maleic anhydride grafting. The highest results were obtained for the polymers tested with primers based on Bondyram 1001-LE polypropylene with the lowest degree of grafting of 0.66% (41.06 N). In turn, for Bondyram 1001 and Bondyram 2101, with grafting degrees of 1.00 and 1.43%, respectively, the results obtained were lower and similar to each other (31.08 and 29.68 N). It shows that the growing polar nature of the polypropylene chain interferes with intermolecular diffusion between the polypropylene substrate and the modified polypropylene in the primer solution.

Polypropylene is a hydrophobic material with a deficient polar component of surface free energy. It does not form any interactions with polar substances but repels them from its surface. In contrast, increasing the degree of grafting of polypropylene with maleic anhydride changes its

hydrophobic nature towards the polar side. This results in differences in the polarity of the two materials. The greater this difference, the more strongly they will repel each other. In turn, the polymer with the lowest degree of grafting, Bondyram 1001-LE, retains sufficient polarity so that molecular diffusion is not significantly inhibited and, on the other hand, allows the formation of covalent bonds with the adhesive.

4. Conclusions

Adhesive joints composed of a polypropylene substrate, a primer layer, and an adhesive tape were subjected to strength analysis. The primer was usually prepared by dissolving the active substance of polypropylene grafted with maleic anhydride in xylene, achieving 0.5% solution. In the tests, the influence of selected parameters on the strength of the joint was checked: primer concentration, the temperature of primer application, type of solvent, time and pressure of the hydraulic press, and grafting degree of polypropylene with maleic anhydride.

The results of the study show that the maximum strength of joints made with primers is achieved at a low concentration of polypropylene grafted maleic anhydride of 0.5 to 1% (average 35.53 N and 37.29 N). At higher concentrations, the strength of adhesive joints decreases with 31.81 N and 30.51 N for 5 and 10 %, respectively, due to the gelation effect that occurs for PP-g-MAH primers with concentrations higher than 1 %. At the same time, chlorinated polyolefins are used from 3 to 20%. Reducing the amount of the active substance in the primer leads to lower production costs, easier preparation of the composition, lower solution viscosity, and the possibility of increasing the contribution of other additives.

The decrease in power is related to the viscosity of the solution, which limits the diffusion of the polymer. In addition, concentrated solutions are applied as a thicker, white layer that weakens the bond. This shows that the modification of polypropylene with maleic anhydride

leads to a significant reduction in its solubility in aromatic hydrocarbons. However, good solubility is a condition that allows for low viscosity and high adhesion.

Primers based on PP-g-MAH without any additives or supporting resins require a high-temperature application. At application temperatures of 25 and 50 °C, the adhesive primer did not show the ability to increase the adhesive properties of the substrate (average sample strength value 16.58 N and 15.91 N, respectively). Only after 80 °C did that joint strength increase by an average of 60%, and at 100 and 120 °C, 117 and 137%, respectively. These results confirm that a diffusion adhesion mechanism forms the connection. The increased temperature and the presence of a solvent confer flexibility to the polymer chains and the ability to create interphase.

According to this theory, the change in solvent also caused significant differences in the adhesive properties of the primer. Higher strength results were obtained using decalin (45.19 N) and toluene (42.79 N), which increased the solubility of both the polypropylene substrate and the active ingredient. In turn, tetrachloroethane and xylene are due to the lower solubility of 37.27 and 35.54 N, respectively. These results show that changing the solvent makes obtaining a higher peel strength possible when the primer is applied at a lower temperature. This conclusion underscores the delicate balance between enhancing reactivity and maintaining substrate compatibility in adhesive interactions. Study results suggest that selecting the appropriate solvent is the key to achieving higher joint strength when the primer is applied at low temperatures.

The results obtained by applying different pressure forces and pressing times show that joints formed using foam adhesive tape require the application of pressure (average 23.5 N without pressure). The adhesive joint should be pressed moderately or briefly, as this reduces its durability. After applying a pressure of 20 bar, the joints were strengthened by an average of 71% (40.22 N). A more powerful compression only leads to a weakening of the connection. The

pressure time should be short since the highest peeling strength (43.93 N) was reached in 10 seconds, while prolonged compression in 60 seconds reduced this value by an average of 23% (33.54 N). This finding indicates that the foam tape is strongly deformed during long compression times or at high-pressure forces, which contributes to the breaking of the contact between the tape and the substrate.

As the degree of grafting of polypropylene with maleic anhydride increased, a decrease in the adhesive properties of the joint was observed. The recommended degree of maleic anhydride grafting is less than 0.7% because, for 0.66% obtained, the highest peel strength result was 41.06 N. For 1.0%, the results were 24.3% (31.08 N) lower, and for 1.43%, a little more than 27.7% (29.68 N). The conclusion states that the increasing polarity of the polymer chain limited intermolecular diffusion into the hydrophobic substrate.

STATEMENT

During the preparation of this work the authors used Omnic 9.2.86 in order to preparation FTIR spectra.

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The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

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Notes

The authors declare no competing financial interest.

Detailed flow chart of the study is attached in support information.

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ABBREVIATIONS

PP-g-MAH, polypropylene grafted maleic anhydride; FTIR, Fourier-transform infrared spectroscopy.

SUPPORTING INFORMATION (SI)

As supporting information, we included particular peel strength records for each experiment.

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Graphical Abstract

